

LIFE CYCLES

An animal's life cycle consists of all the stages between the beginning of one generation and the beginning of the next. In some species—especially insects and other small invertebrates—the entire cycle is completed within a few weeks; in much larger animals, it often takes many years. Regardless of how long it takes to complete, an animal's life cycle always involves 2 main steps: a period of growth and development, followed by reproduction. Some animals reproduce once and then die: for them, reproduction marks the end of life as well as the completion of the life cycle. For many, reproduction continues throughout adulthood, giving animals more than one chance to produce young.

Reproduction

The ability to reproduce is the cornerstone of life because it allows living things to multiply, exploit new opportunities around them, and evolve. Animals reproduce in one of 2 ways: asexually (without sex) or sexually.

In asexual reproduction, a single parent partitions off part of itself to form a new animal. The partitioning process can happen in a variety of ways. Hydras, for

SEXUAL REPRODUCTION

As a female common frog lays her eggs, her mate sprays them with sperm to fertilize them. Each tadpole will be genetically unique.



ASEXUAL REPRODUCTION

This sea anemone is in the final stages of reproducing, which it does by tearing itself in half. The result will be 2 individuals that have exactly the same genetic makeup.

LIFESPANS

In general, animal lifespans are directly related to adult body size: the larger the animal, the longer it is likely to live, although there are exceptions to this rule. One important factor affecting lifespan is metabolic rate—the rate at which an animal uses energy to make its body work. “Cold-blooded” animals, such as amphibians and reptiles, have a relatively low metabolic rate and tend to be long-lived, while “warm-blooded” animals, such as birds and mammals, have a high metabolic rate and tend to be relatively short-lived. This is especially true of small species because a low body mass means that heat escapes quickly from the body and has to be constantly replaced by food. Environmental factors, such as temperature and humidity, also have an effect. Houseflies, for example, often die within 6 weeks in warm conditions but may survive for many months if it is cool, and some microscopic animals survive for decades if they remain in a dormant state. In general, animals rarely live for long after their reproductive life has come to an end.

example, produce small buds that grow into complete new animals, while sea anemones literally tear themselves in 2. Some animals produce eggs that develop without fertilization—a process called parthenogenesis. This is common in aphids and other sap-sucking insects, but it is rare in vertebrates (whiptail lizards are one of the few examples). Asexual reproduction is relatively quick and simple, but it has one important disadvantage: since only one parent is involved, the offspring are either genetically identical to that parent or very similar to it. As a result, parent and offspring are equally vulnerable to threats such as disease: if one animal dies, the rest will often follow suit.

Sexual reproduction gets around this problem because the involvement of 2 parents produces offspring that are genetically varied. Each one has a unique combination of characteristics, which allows the fittest to survive and the species slowly evolves. However, the disadvantage of sexual reproduction is that it is much more complicated: the parents must be of the right species and correct sex, and in most cases they must cooperate to breed. In addition, only one parent—the female—actually produces young, so some reproductive potential is lost.

Despite these difficulties, sexual reproduction is widespread throughout the animal world, which demonstrates its long-term value. Even species that normally reproduce without sex periodically include a sexual phase in their life cycle, thereby getting the best of both worlds.

Animal sexes

All animals that reproduce sexually with a partner of the opposite sex show dimorphism—that is, the males and females are anatomically different and behave in different ways. In some species, the differences are not obvious, but in others, they are quite distinct. Dimorphism exists because the sexes have developed different roles in reproduction and need different body forms to carry them out.

However, not all sexually reproducing animals are of opposite sexes. Some—earthworms and terrestrial snails, for example—are hermaphrodite



DIMORPHIC PARTNERS

In some animals, sex differences are extreme. Here, a wingless female vapourer moth (seen on the right) has attracted a winged male. After mating, the female will crawl away to lay her eggs.

(they have both male and female sex organs). This simplifies sexual reproduction because any adult of a species is potentially a suitable mate for another. A further variation is that some species have separate sexes but individuals can change sex during adult life. Parrotfishes, for example, often live in small schools dominated by a single male. If the male dies, a female changes sex and takes his place.

Courtship

Before an animal can mate, it has to find a partner. This is easy enough for species that live in groups, but for animals that live alone, it poses problems. Solitary animals locate potential mates by sending out signals, such as sounds or airborne scents. Each species has its own characteristic “call sign,” ensuring that it finds others of its own kind.

Once the sexes are in contact, one partner—usually the male—has to overcome the other's wariness and demonstrate his suitability as a mate. This process is known as courtship. It often takes the form of ritualized behavior that displays the male's physical fitness or his ownership of a good provision of food. If the female is sufficiently impressed, she will accept him as her mate.

While some species form lifelong partnerships, many go their separate ways after mating. In the latter case, the males typically mate with several females but take no part in raising the young. More rarely, things work in the opposite way, with one female mating with several males. Where this happens—for example, in phalaropes (see p.312)—the female is more brightly colored than the male and often takes the lead in courtship. In general, these females take little or no part in rearing young.

BODY SIZE



shrew (up to 4 in/10 cm): 1 year



elephant (up to 25 ft/7.5 m): 70 years

WARM-BLOODED AND COLD-BLOODED



rock pigeon (warm-blooded): 10 years



common toad (cold-blooded): 35 years

ENVIRONMENTAL CONDITIONS



housefly (at 77°F/25°C): 6 weeks

housefly (at 59°F/15°C): 18 weeks

FACTORS DETERMINING LIFESPAN

Three of the major factors influencing an animal's lifespan are illustrated above. The life expectancy of each animal is represented by a colored bar, indicating the degree to which a factor can affect its lifespan.